

Q1. Define the following terms:

- (a) open-loop control system,
- (b) closed-loop control system,
- (c) feedback,
- (d) transfer function.

[2 Marks]

Q2. Explain the advantages and disadvantages of a closed-loop control system compared with an open-loop control system. [3 Marks]

Q3. Obtain the transfer function of the system shown in the given block diagram. Clearly indicate all block-diagram reduction steps. [5 Marks]

Q4. Consider a system governed by the differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 2u(t).$$

Assuming zero initial conditions, determine the transfer function

$$G(s) = \frac{Y(s)}{U(s)}.$$

Also determine the poles and zeros of the system.

[5 Marks]

Q5. Represent the following transfer function in state-space form:

$$G(s) = \frac{2s + 5}{s^2 + 4s + 3}.$$

[5 Marks]

Additional Practice Questions

- P1.** Draw the block diagram of a standard negative-feedback control system and identify its major components.
- P2.** Determine the overall transfer function of a negative-feedback system having forward-path transfer function $G(s)$ and feedback-path transfer function $H(s)$.
- P3.** Explain the physical significance of poles and zeros in a linear control system.
- P4.** Differentiate between transfer-function and state-space representations.

Student Remarks or Difficulties Encountered

Faculty Evaluation

Marks Obtained	
Faculty Remarks	
Faculty Signature	

Department of Electrical and Electronics Engineering